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#include<iostream>
#include<GL/glut.h>
#include<stdlib.h>
#include<math.h>
using namespace std;
int v, wxmin, wxmax, wymin, wymax, a[10][2];
void init(void)
{
glClearColor(0.0,0.0,0.0,0.0);
gluOrtho2D(-600.0,600.0,-600.0,600.0);
}
int accept_poly(int a[10][2])
{
int i,n;
cout<<"\nEnter the number of vertices : ";
cin>>n;
for(i=0;i<n;i++)
{
cout<<"Enter the coordinates"<<i + 1 << " : " ;
cin>>a[i][0]>>a[i][1];
}
return n;
}
int leftclip(int a[10][2],int v,int wxmin)
{
int i,t[10][2];
int k=0;
int x1,y1,x2,y2;
for(i=0;i<v;i++)
{
if(a[i][0] >= wxmin)
{
t[k][0]=a[i][0];
t[k][1]=a[i][1];
k++;
}
if((a[i][0] > wxmin && a[(i+1)%v][0] < wxmin) || (a[i][0] < wxmin &&
a[(i+1)%v][0] > wxmin))
{
x1=a[i][0];
y1=a[i][1];
x2=a[(i+1)%v][0];
y2=a[(i+1)%v][1];
t[k][1] = y1 + (wxmin-x1) * ((float)(y2-y1)/(x2-x1));
t[k][0] = wxmin;
k++;
}
}
for(i=0;i<k;i++)

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{
a[i][0] = t[i][0];
a[i][1] = t[i][1];
}
return(k);
}
int rightclip(int a[10][2],int v,int wxmin)
{
int i,t[10][2];
int k=0;
int x1,y1,x2,y2;
for(i=0;i<v;i++)
{
if(a[i][0] <= wxmax)
{
t[k][0]=a[i][0];
t[k][1]=a[i][1];
k++;
}
if((a[i][0] < wxmax && a[(i+1)%v][0] > wxmax) || (a[i][0] > wxmax
&& a[(i+1)%v][0] < wxmax))
{
x1=a[i][0];
y1=a[i][1];
x2=a[(i+1)%v][0];
y2=a[(i+1)%v][1];
t[k][1] = y1 + (wxmax-x1) * ((float)(y2-y1)/(x2-x1));
t[k][0] = wxmax;
k++;
}
}
for(i=0;i<k;i++)
{
a[i][0] = t[i][0];
a[i][1] = t[i][1];
}
return(k);
}
int topclip(int a[10][2],int v,int wymax)
{
int i,t[10][2];
int k=0;
int x1,y1,x2,y2;
for(i=0;i<v;i++)
{
if(a[i][1] <= wymax)
{
t[k][0]=a[i][0];
t[k][1]=a[i][1];

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k++;
}
if((a[i][1] < wymax && a[(i+1)%v][1] > wymax) || (a[i][1] > wymax
&& a[(i+1)%v][1] < wymax))
{
x1=a[i][0];
y1=a[i][1];
x2=a[(i+1)%v][0];
y2=a[(i+1)%v][1];
t[k][1] = wymax;
t[k][0] = x1 + (wymax-y1) * ((float)(x2-x1)/(y2-y1));
k++;
}
}
for(i=0;i<k;i++)
{
a[i][0] = t[i][0];
a[i][1] = t[i][1];
}
return(k);
}
int bottomclip(int a[10][2],int v,int wymin)
{
int i,t[10][2];
int k=0;
int x1,y1,x2,y2;
for(i=0;i<v;i++)
{
if(a[i][1] >= wymin)
{
t[k][0]=a[i][0];
t[k][1]=a[i][1];
k++;
}
if((a[i][1] > wymin && a[(i+1)%v][1] < wymin) || (a[i][1] < wymin &&
a[(i+1)%v][1] > wymin))
{
x1=a[i][0];
y1=a[i][1];
x2=a[(i+1)%v][0];
y2=a[(i+1)%v][1];
t[k][1] = wymin;
t[k][0] = x1 + (wymin-y1) * ((float)(x2-x1)/(y2-y1));
k++;
}
}
}
for(i=0;i<k;i++)
{
a[i][0] = t[i][0];

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a[i][1] = t[i][1];
}
return(k);
}
void draw_poly(int a[10][2],int v)
{
int i;
glBegin(GL_LINES);
glVertex2i(wxmin,wymin);
glVertex2i(wxmin,wymax);
glVertex2i(wxmin,wymax);
glVertex2i(wxmax,wymax);
glVertex2i(wxmax,wymax);
glVertex2i(wxmax,wymin);
glVertex2i(wxmax,wymin);
glVertex2i(wxmin,wymin);
glEnd();
glFlush();
for(i=0;i<v;i++)
{
glColor3f(1.0,1.0,1.0);
glBegin(GL_LINES);
glVertex2i(a[i][0],a[i][1]);
glVertex2i(a[(i+1)%v][0],a[(i+1)%v][1]);
glEnd();
glFlush();
}
}
int main(int argc,char** argv)
{
int cl;
char ans;
cout<<"\nEnter the co-ordinates of the window";
cout<<"\n(Wxmin, Wymin) : ";
cin>>wxmin >> wymin;
cout<<"\n(Wxmax, Wymax) : ";
cin>>wxmax >> wymax;
v = accept_poly(a);
glutInit(&argc,argv);
glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
glutInitWindowSize(800,800);
glutInitWindowPosition(100,100);
glutCreateWindow("Clipping");
init();
glClear(GL_COLOR_BUFFER_BIT);
draw_poly(a,v);
glFlush();
do
{

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cout<<"\nWhich clipping do you want to perform??\n1. Left\n2. Right\n3. Top\n4.Bottom\nEnter your choice : ";
cin>>cl;
switch(cl)
{
case 1 :
glClear(GL_COLOR_BUFFER_BIT);
v=leftclip(a,v,wxmin);
glClear(GL_COLOR_BUFFER_BIT);
draw_poly(a,v);
break;
case 2 :
glClear(GL_COLOR_BUFFER_BIT);
v=rightclip(a,v,wxmax);
glClear(GL_COLOR_BUFFER_BIT);
draw_poly(a,v);
break;
case 3 :
glClear(GL_COLOR_BUFFER_BIT);
v=topclip(a,v,wymax);
glClear(GL_COLOR_BUFFER_BIT);
draw_poly(a,v);
break;
case 4 :
glClear(GL_COLOR_BUFFER_BIT);
v=bottomclip(a,v,wymmin);
glClear(GL_COLOR_BUFFER_BIT);
draw_poly(a,v);
break;
}
cout<<"\nDo you want to perform another clipping?? (Y/N) ";
cin>>ans;
}while(ans=='y' || ans=='Y');
glFlush();
glutMainLoop();
return 0;
}

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